

Method, Then Measurement

DATA 443/643 · Week 15, note 15.1

Week 15

American University

Ahmad Mousavi

A recap of DATA 443/643. Method then measurement. Original slides.

Lecture plan

- 1 Four modules, one stack
- 2 A formula sheet you can write
- 3 Failure → the number that catches it
- 4 Remaining talks: Canvas + a failure case

Key goal: name the method, then the measurement. Trivia is not the exam.

First time: what / why / formula

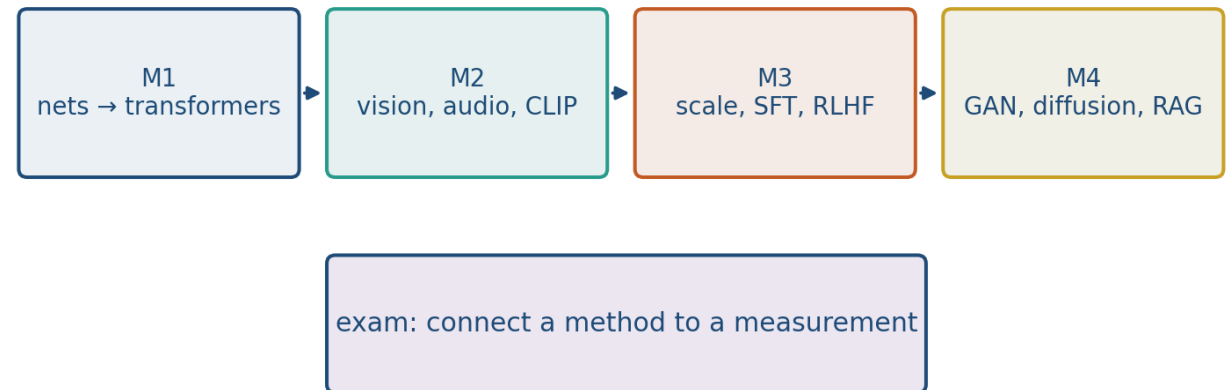
- What / why. Four modules. You name the method, then the number that would have caught the failure. Fluency is not a check. RAG without recall@k, GANs without coverage, CoT without faithfulness are incomplete.
- Architecture. Closed-book written exam plus remaining talks. Formula sheet is what you can write, not trivia.
- Formula. Write the method's equation, then the measurement (ASR, recall@k, $D^* = 1 / 2$, $\sigma(0) = 1 / 2$, ...).
- Tradeoffs. + Cumulative and fair if you studied the first-time blocks. – Naming a paper without a metric scores poorly.

How. Practice: "same face, low loss_G " → mode bins. "Cite [d5], fact in [d1]" → attribution + recall@k.

What sits on the exam

- M1 nets → transformers. M2 vision, audio, CLIP.
- M3 scale, SFT, RLHF. M4 GAN, diffusion, RAG.
- Draw the picture. Attach one number.

Four modules, one stack. The project sits on top of all four.



Write these, then say what you would plot

Attention / GAN

$$\text{softmax} (QK^\top / \sqrt{d})V.$$
$$D^* = p_r / (p_r + p_g).$$

Diffusion / CFG

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon.$$
$$\hat{\varepsilon} = \varepsilon_u + s(\varepsilon_c - \varepsilon_u).$$

DPO / cosine

$$\sigma(0) = 1 / 2.$$
$$\cos(a, b) = a^\top b / (\|a\| \|b\|).$$

The exam shape

- Same face, low lossG → mode bins, not lossG.
- Right box, last step 52 → F among wins.
- Cite [d5], fact in [d1] → attribution + recall@k.

Exam shape: name the method, then the number that would have caught it.

failure	measure
same face, low lossG	mode bins
right box, last step 52	faithful among wins
cite [d5], fact in [d1]	attribution + recall@k

Pairs the exam likes

Architecture choice

GPT vs BERT. CLIP vs BLIP. GAN vs diffusion. CoT vs ReAct. Justify with the task.

The metric

Then say what you would plot. Labs 11–14 are the Module 4 shortcuts.

What to take from this lecture

Labs on paper

11 bins, 12 oracle reverse, 13 last-int vs box, 14 cosine + calc.

Talks

Canvas is official. Bring a failure, not only the best run.

Karpathy

Recap of the LLM stack. Not a substitute for Weeks 11–12.